

Understand → Recall → Practise → Revise

1

Electric charge

Distinguish charge types and explain conservation and quantisation.

p. 1

- Two types: positive and negative.
- Like charges repel; unlike charges attract.
- SI unit: coulomb (C).

Self-check: What happens when two positive charges are brought near each other?

2

Coulomb's law

Use the inverse-square relation and identify force direction.

p. 2

- $F = k |q_1 q_2| / r^2$.
- Force acts along the line joining the charges.
- Double $r \rightarrow$ force becomes $F/4$.

Self-check: What happens to force when separation doubles?

3

Electric field

Separate field from force and interpret field-line direction.

p. 3

- Vector quantity: magnitude and direction.
- SI unit: N/C.
- Use a small test charge to avoid disturbing source charges.

Self-check: Why should a test charge be small?

4

Electric potential

Distinguish potential, potential difference and electric field.

p. 4

- Scalar quantity: no vector direction.
- SI unit: volt = joule/coulomb.
- Potential difference is change in potential energy per unit charge between two points.

Self-check: Is electric potential a vector?